

CS316: INTRODUCTION TO AI  
INSTRUCTOR: DR.SAWSAN ALHALAWANI

# PREDICTING ELECTRIC VEHICLE CHARGING STATIONS DEMAND

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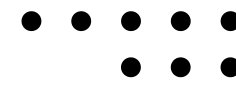
# INTRODUCTION

Electric vehicles are becoming increasingly popular, placing greater pressure on public charging stations. The usage of these stations changes constantly depending on the time of day, weather conditions, and traffic levels, which makes demand difficult to predict.

This unpredictability causes long waiting times, energy inefficiencies, and poor user experience.

In this project, we develop a machine-learning system that analyzes historical charging data to forecast future demand and support better planning for energy distribution.





# MOTIVATION

Public charging stations often struggle with sudden demand spikes that lead to congestion and unnecessary delays. By accurately forecasting charging behavior, operators can prepare for peak hours, balance the energy load, and reduce pressure on the grid.

Effective forecasting also improves user experience and supports the transition toward sustainable smart cities.

Our motivation is to create a prediction model that helps optimize charging operations and ensures more efficient, reliable EV infrastructure.



# DATASET OVERVIEW

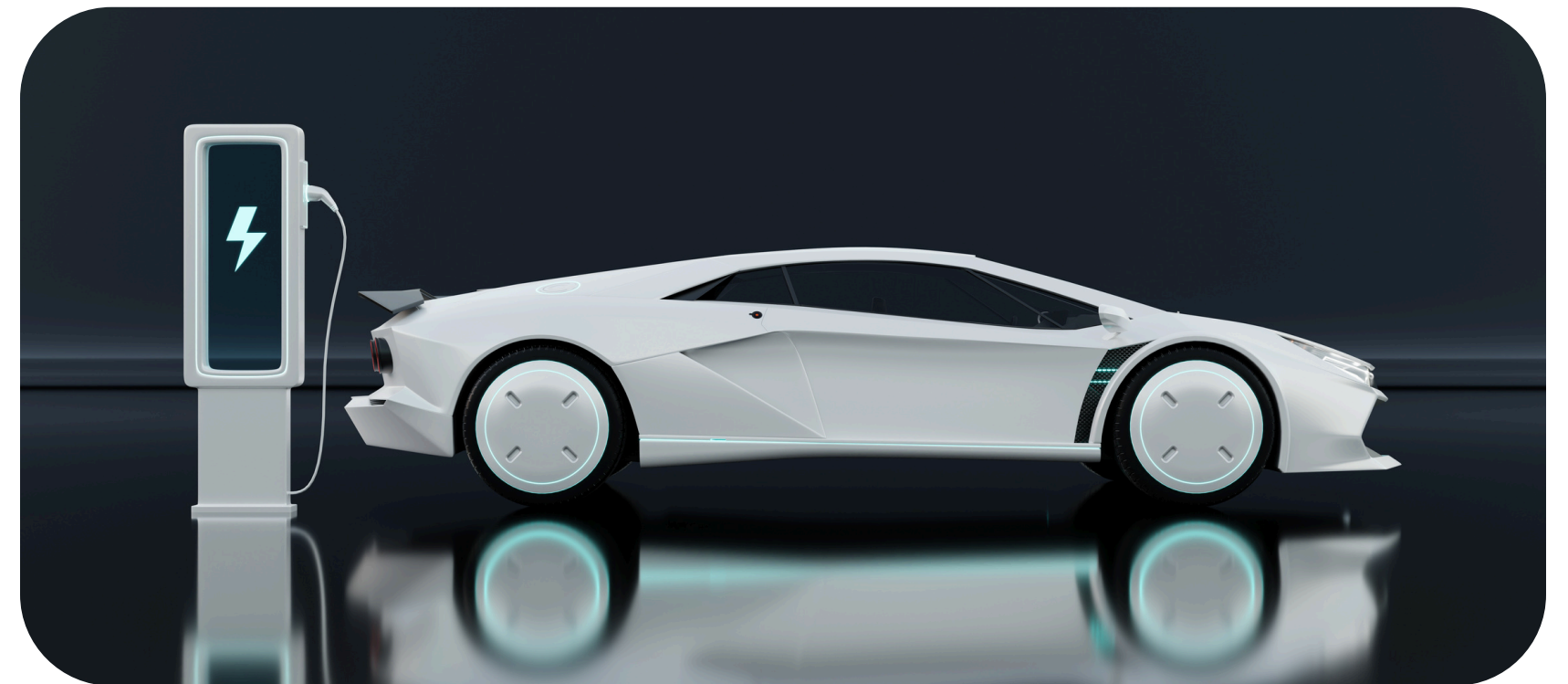


Our project uses the “Charging Pile Demand Forecasting” dataset from Kaggle, which contains real-world information from electric vehicle charging stations.

## **The dataset includes:**

- Timestamp (date/time of charging)
- Charging duration
- Energy consumed
- Number of EVs charging
- Traffic flow
- Peak traffic hour indicators
- Vehicle type
- Temperature

These operational, environmental, and temporal features make it possible to analyze user behavior and predict charging demand with high accuracy.



# NEW DERIVED FEATURES

To improve model accuracy, we created new features:



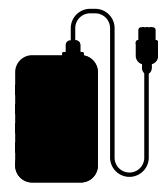
## 1. Duration per EV

- Average charging time per vehicle
- Helps identify slow vs. fast charging periods



## 3. Rolling Usage Rate (Moving Average)

- Smooths short-term fluctuations
- Captures weekly/hourly demand patterns



## 2. Energy per Minute

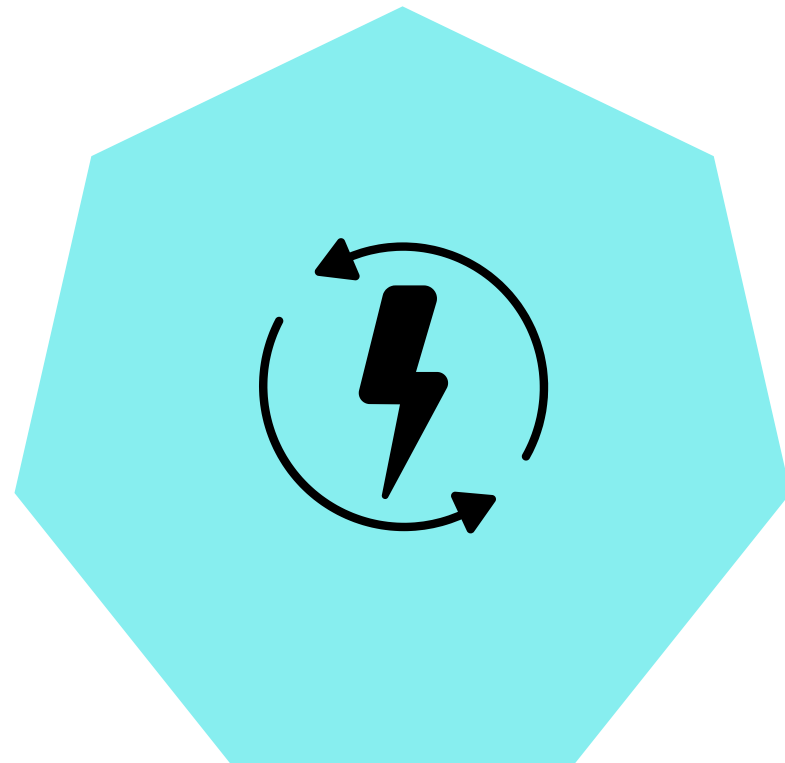
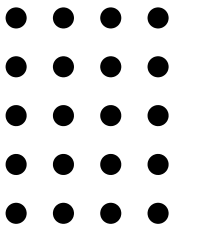
- Rate of energy consumption
- Indicates station efficiency and load



## 4. Interaction Features

- (e.g., temperature  $\times$  hour of day)
- Helps models detect non-linear behavior caused by weather and time

# FEATURE ENGINEERING



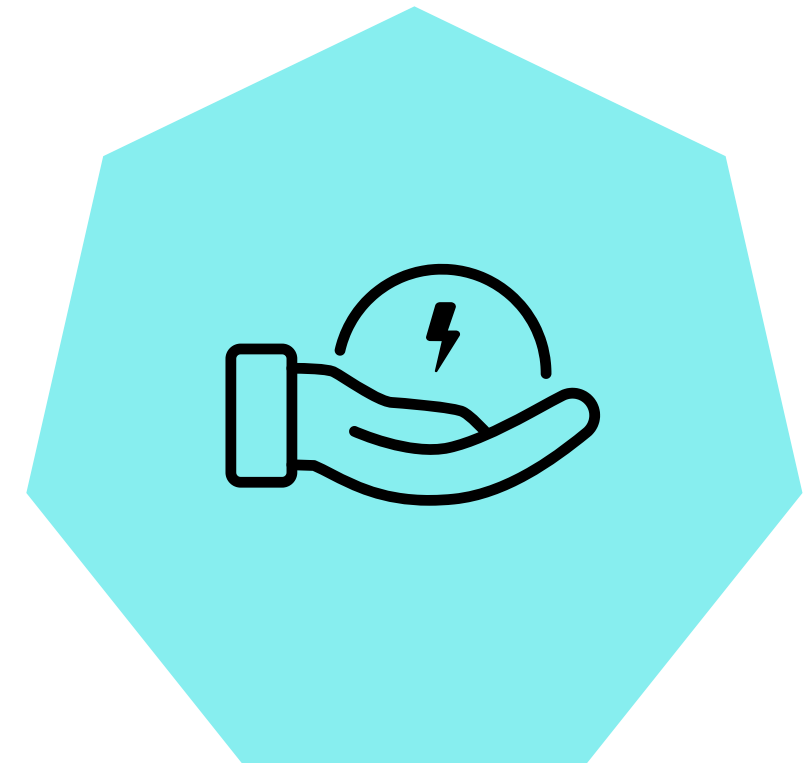
## **DURATION PER EV**

Operates only on stored electrical energy; produces no direct emissions.



## **ENERGY PER MIN**

Combines an electric motor with an alternative power source for flexibility.

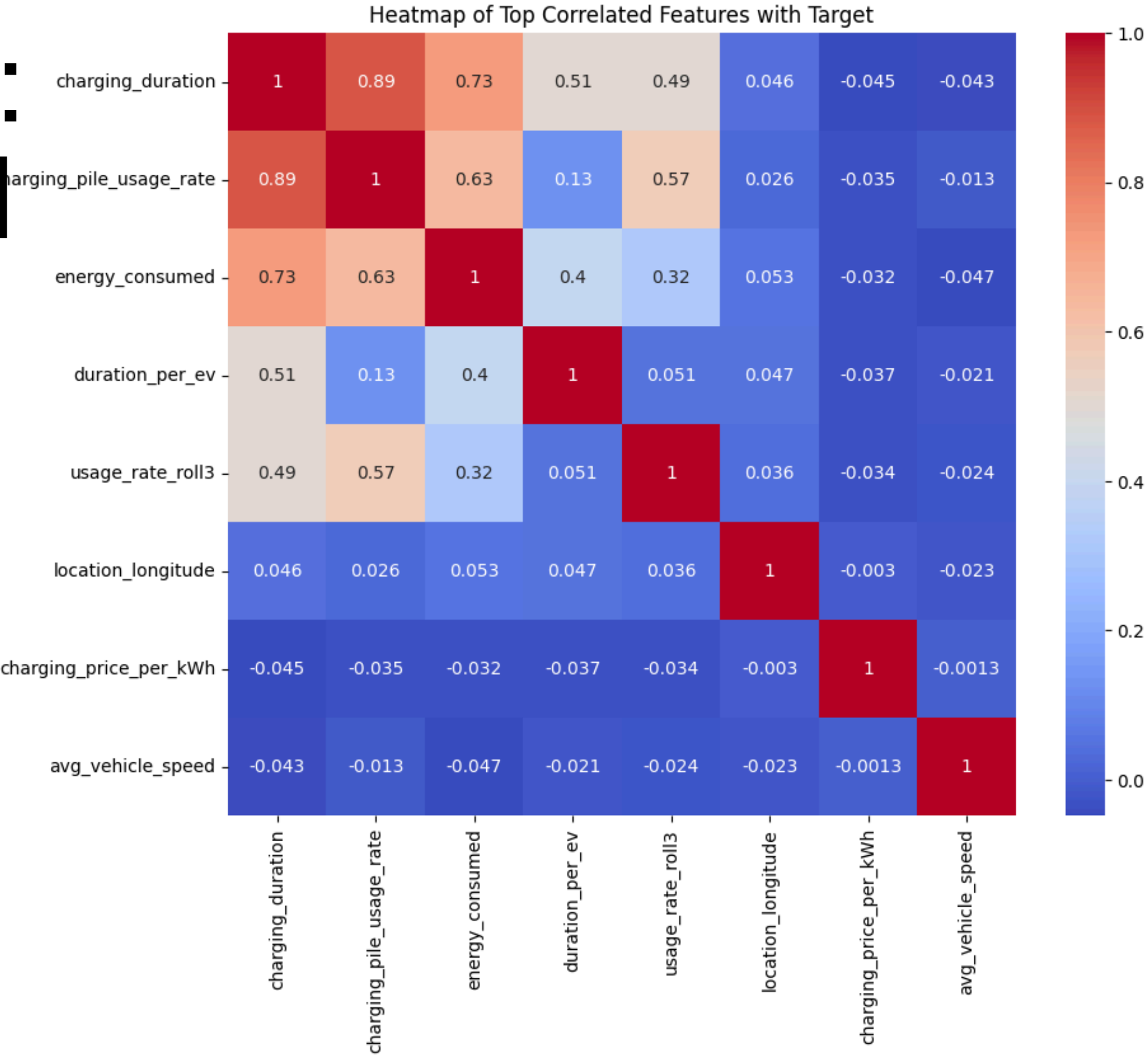


## **ROLLING USAGE RATE**

Uses a small generator to recharge the battery while driving for longer distance.

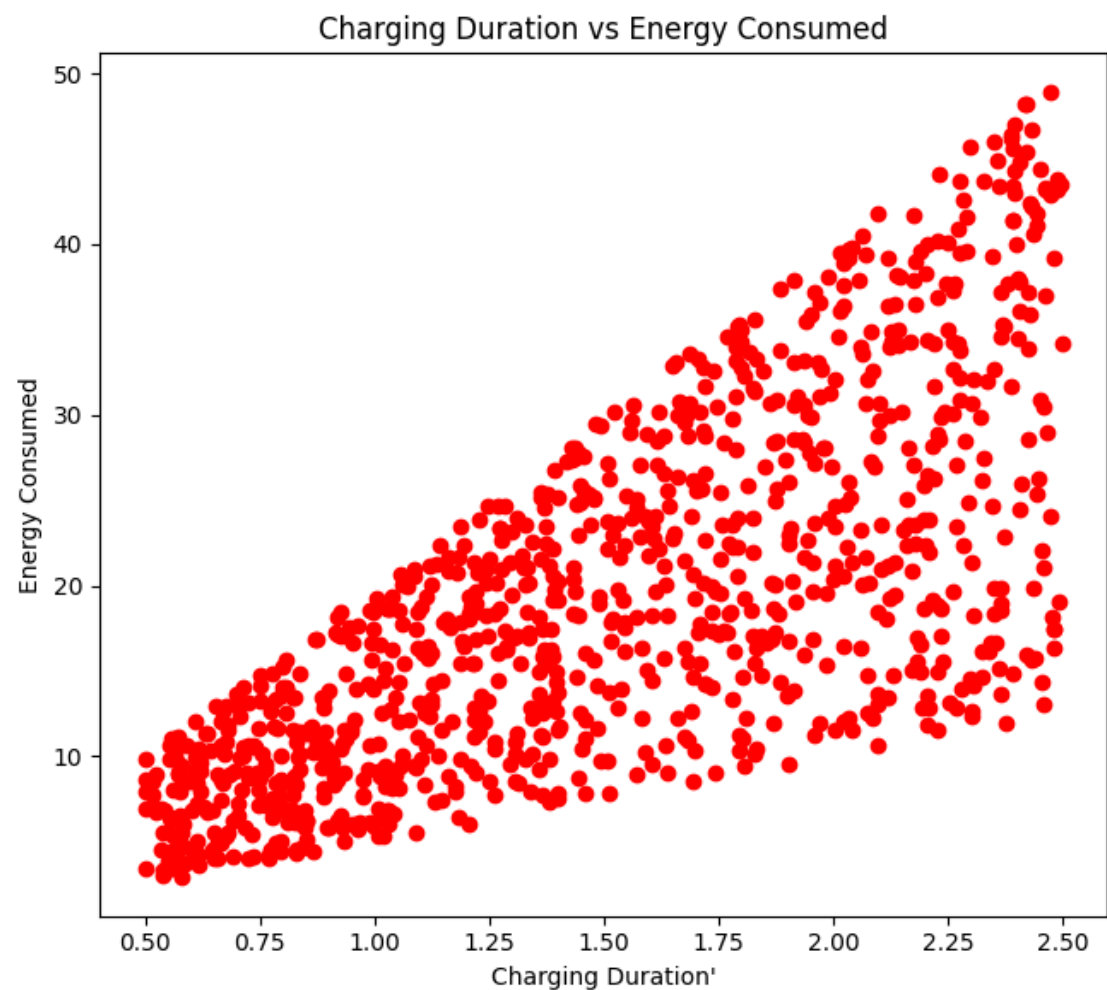
# KEY FINDINGS: CORRELATION HEAT MAP

- Charging Duration and Energy Consumed have **Strong Correlation** with Usage Rate.
- **Weak correlations** with Weather variables.
- Traffic-related features show **moderate influence**.
- Engineered features (duration\_per\_ev, energy\_per\_min, usage\_rate\_roll3) add **predictive strength**.

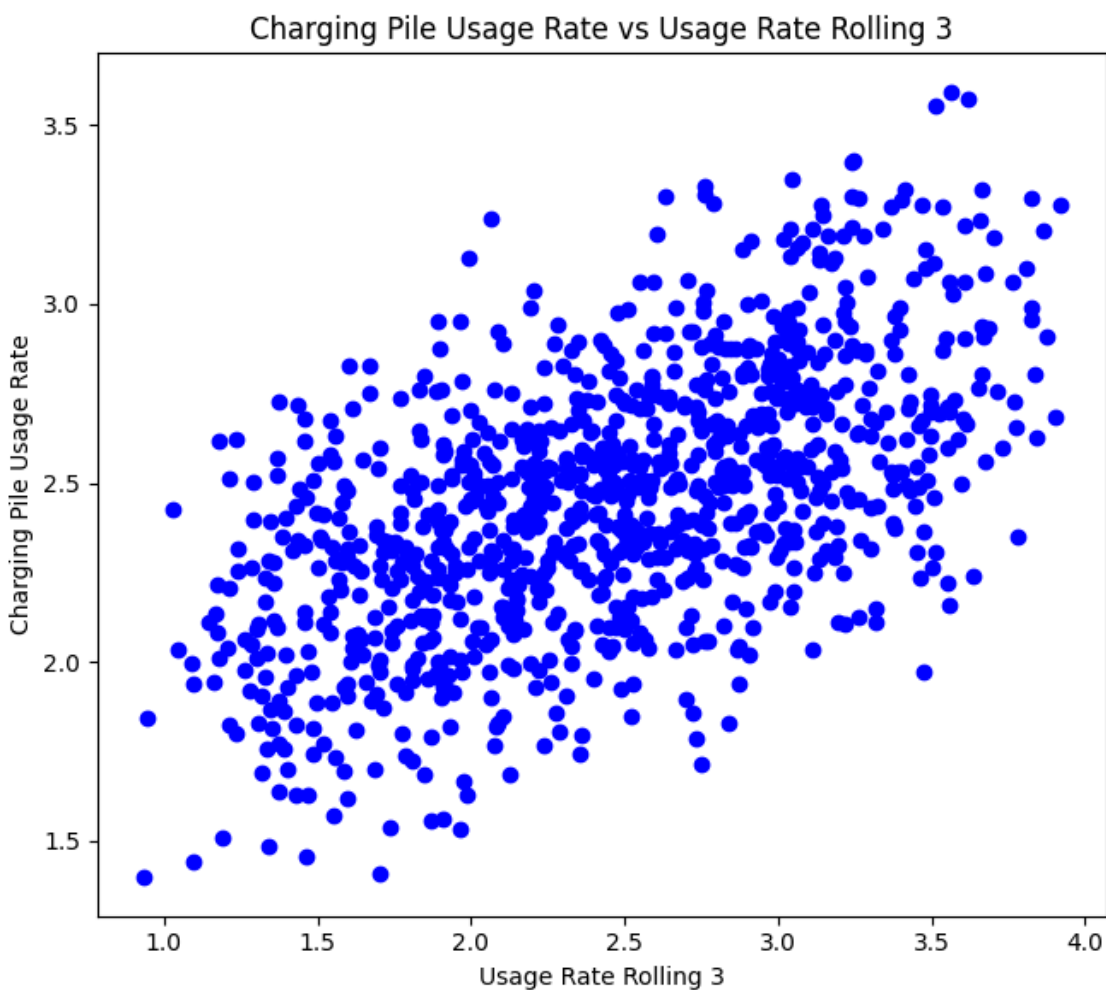


# KEY FINDINGS: FEATURE RELATIONSHIPS

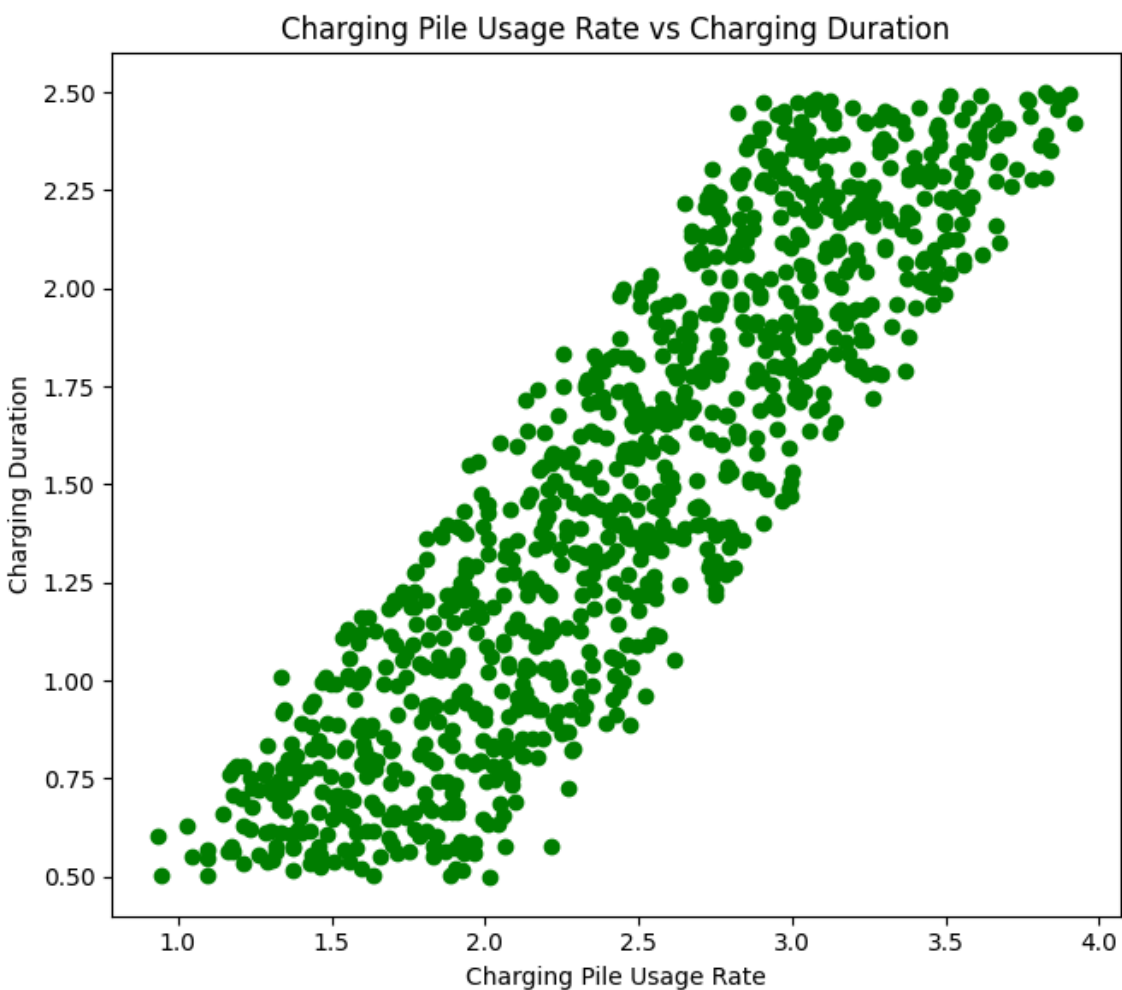
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Highly Strong Positive Relationship

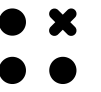


Moderately Strong Positive Relationship

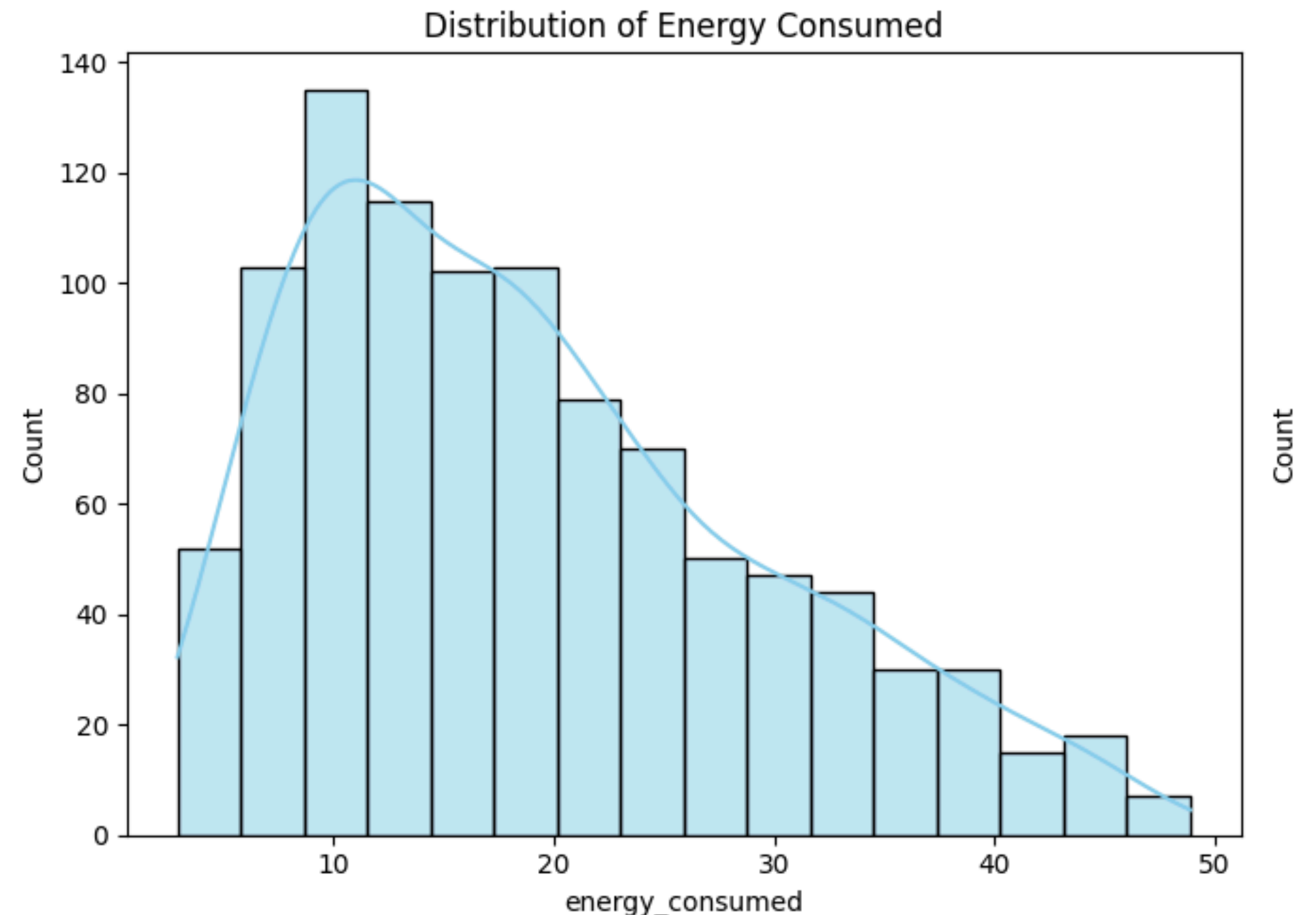


Strong Positive Relationship

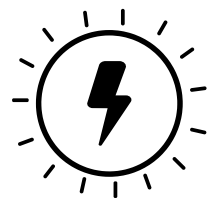
# KEY FINDINGS: DISTRIBUTION



- **Right-skewed;** most charging sessions consume a moderate amount of energy, while very high-energy sessions are rare.
- Typical charging sessions consume between ~10 to 25 kWh, representing normal daily charging rather than full battery cycles.
- Few users perform large charging sessions, which may correspond to long-distance travel or low initial battery levels.

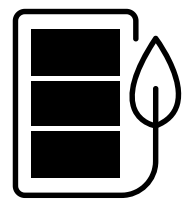


# MODELS USED



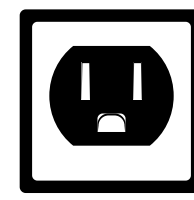
## Linear Regression

Simplest baseline model, fast and simple



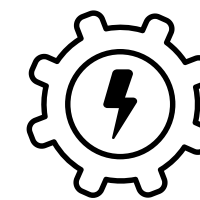
## Decision Tree

Captures nonlinear patterns, interpretable



## Random Forest

More accurate ensemble model, reduces overfitting



## XGBoost

Handles complex relationships, gradient boosting algorithm

# MODEL PERFORMANCE

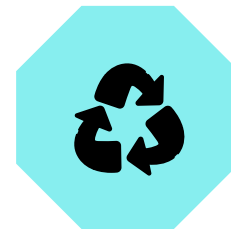
|   | Model             | MAE                  | RMSE                 | R <sup>2</sup> |
|---|-------------------|----------------------|----------------------|----------------|
| 0 | Random Forest     | 0.073                | 0.0898               | 0.9849         |
| 1 | Linear Regression | 0.000000000000000007 | 0.000000000000000009 | 1              |
| 2 | Decision Tree     | 0.1023               | 0.132                | 0.9675         |
| 3 | XGBoost           | 0.0565               | 0.0722               | 0.9903         |

# MAIN INSIGHTS



## **Charging Duration = Strongest Predictor**

Major factor influencing congestion.



## **Peak Usage Hours = High Traffic**

Shows clear demand patterns.



## **Weather Factors Have Little Impact on Charging Behavior**

Minimal influence compared to operational and traffic features.

## **Strong Correlation Between Energy Consumption & Usage Rate**

indicating predictable charging behavior.

## **XGBoost Had Best Performance**

Most reliable for forecasting charging demand.

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# THANK YOU FOR YOUR ATTENTION

Together, let's imagine a cleaner and smarter future of mobility

